MAHYAR JABERI

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EXPERIENCE

AI Solutions Engineer Co-op Aug 2026 – Present
FGF Brands *Vaughan, ON*

- Cut LLM cost 81% and build time 56% for an AI app-builder agent on benchmarks (114→32 model calls, 8/8 changes still correct) by profiling its tool loop and adding diff-returning batched edits and cache-stable prompts
- Built a Dev→UAT→Prod promotion pipeline (FastAPI, React) so LLM agents move between environments instead of being re-prompted in each, with dry-run previews and paused imports (486 checks, 0 secrets exported)
- Added live web search to Claude and GPT chat agents through one provider-agnostic tool on Azure OpenAI, with end-to-end URL citations and prompt-injection guards (untrusted-content tags, per-turn caps)
- Hardened agent tool-calling against excessive agency: shared AI apps and their embedded assistants now run on each viewer's own credentials and only within their manifest's API grants (376 mutation-tested checks)

Undergraduate Researcher (Directed Studies) Sep 2026 – Present
York University, Data Mining Lab (Prof. Manos Papagelis) *Toronto, ON*

- Developed a graph-traversal retrieval tool for a tool-calling LLM agent over 928K+ FHIR records (Python, BLAZE, MIMIC-IV), and patched 2 retrieval bugs in the FHIR-AgentBench harness after auditing it (13 issues)
- Ran 1,489 agent episodes under a pre-registered plan (paired sign-flip tests): the tool alone gave no significant gain, but tracing failures to a schema gap let one prompt line lift correctness 25/84 to 56/84 (exploratory)

Software Developer Intern May 2025 – Aug 2025
Nova Ventures *Toronto, ON*

- Automated parsing of club PDFs and spreadsheets into structured tables, and built FastAPI endpoints and React heatmaps that surfaced per-session goalkeeper performance to coaches

App Developer & Database Intern May 2024 – Aug 2024
Sepantech *Sweden*

- Cut average response time 30% on a course platform serving 500+ medical professionals by optimizing 12+ MySQL queries, and built 8+ Node.js API endpoints for enrollment and certification

PROJECTS

BowlWise (bowlwise.app) Python, FastAPI, Claude API, MCP SDK, MongoDB Atlas, React

- Launched a full-stack dog food recommendation platform, live on customer tablets in 2 Pet Valu stores, that scores all 265 foods for a dog in under 20 ms (p95); 208 dog profiles created to date
- Built Ask BowlWise, a Claude agent with 15 tools over the scoring engine, so scores and prices come from the engine, not the model, and health data never reaches it; passed 111/111 eval conversations at $0.009 each
- Shipped a 7-tool read-only MCP server listed in Claude's connector directory, with required attribution and dated prices, at zero model cost to BowlWise; 30/30 scripted Claude answers passed its checks

Landline Python, FastAPI, Higgs Realtime, Twilio, Next.js, MapLibre

- Placed 3rd of 30+ teams at the Boson AI Higgs Audio Hackathon (Georgian Build North finalist) on a 3-person team building a voice agent that phones Toronto listing agents in parallel and books viewings
- Rebuilt the live-map web app and real-time read API, and fixed the agent's tool use: in evals, sessions where it skipped its tools fell 7/21→0/21 and clean multilingual call endings rose 10/15→15/15

EDUCATION AND CERTIFICATES

Honours BA, Computer Science (Co-op), York University Sep 2023 – Apr 2028 (Expected)
Machine Learning Specialization, Stanford University & DeepLearning.AI In Progress

SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Java, C
AI / LLM	Claude API, OpenAI API, Azure OpenAI, MCP, LangGraph, RAG, tool calling, LLM evals
Frameworks	FastAPI, Pydantic, Node.js, React, Next.js
Data & Cloud	PostgreSQL, MongoDB Atlas, Azure, GCP, Docker, GitHub Actions, Vercel